Order Statistics

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CPE 231 Algorithms

Topics

Selection problem

Lomuto partitioning

Quickselect algorithm

Application Scenario: Web Monitoring Service

When monitoring a web service, we record response times (latency) for thousands or millions of API requests.

Engineers care about high percentiles.

> ***SLA**: 95% of requests should complete within T milliseconds. Given one million latency measurements, is the SLA kept?*



Ex: 20 API request latencies

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12, 15, 11, 18, 21, 14, 13, 18, 16, 17,
12, 15, 19, 14, 13, 16, 18, 22, 11, 12
```

For $0 \leq p \leq 1$,

$$100p^{th} \text{ percentile location } = \lfloor (n-1)p + 1 \rfloor$$

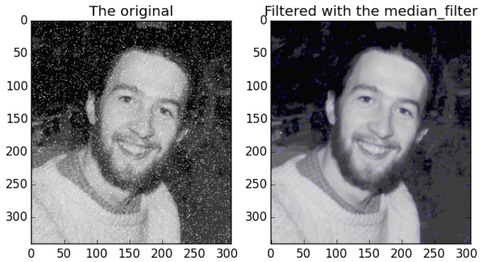
- For $n = 20$ recorded values, what is the location of the 95th Percentile (P95) in the data?
- How would you find it ?

The Selection Problem

Find k^{th} smallest element, or *k^{th} order statistic*, in a list of n numbers

- $k = 1$: Smallest element
- $k = n$: Largest element
- $k = \text{ceil}(n/2)$: Median





Median filter to smooth out noise.

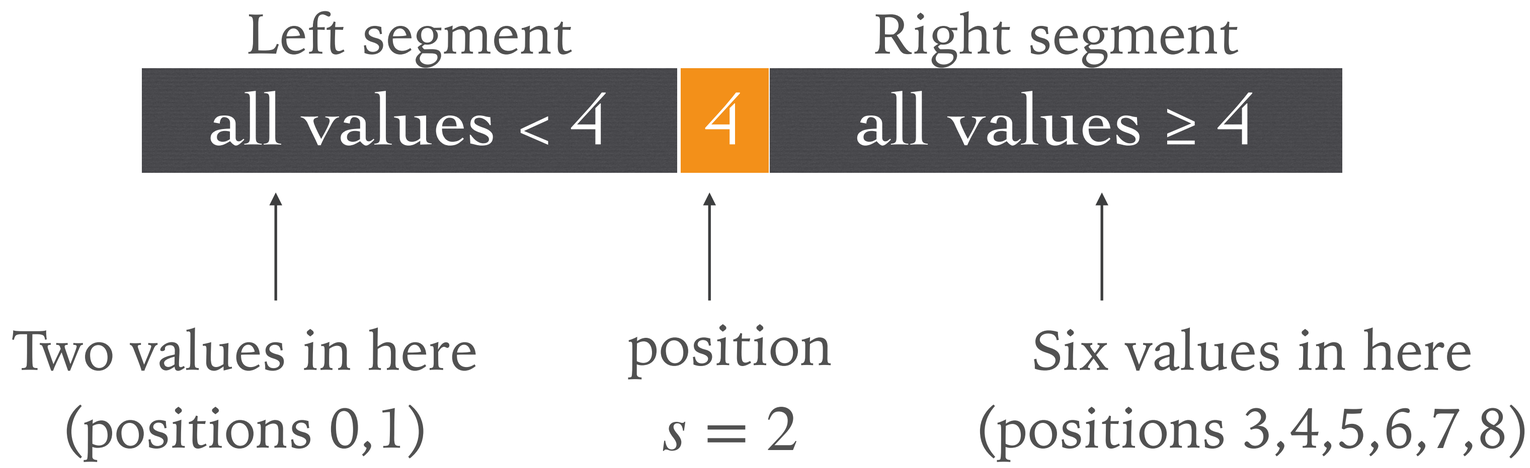
4 1 10 8 7 12 9 2 15

How would you find
- Median ?
- 2nd smallest element ?
- 5th smallest element ?

Choose "4" as the ***pivot element*** to partition the list into three segments:

What is the 5th smallest element (5th order statistic) ?

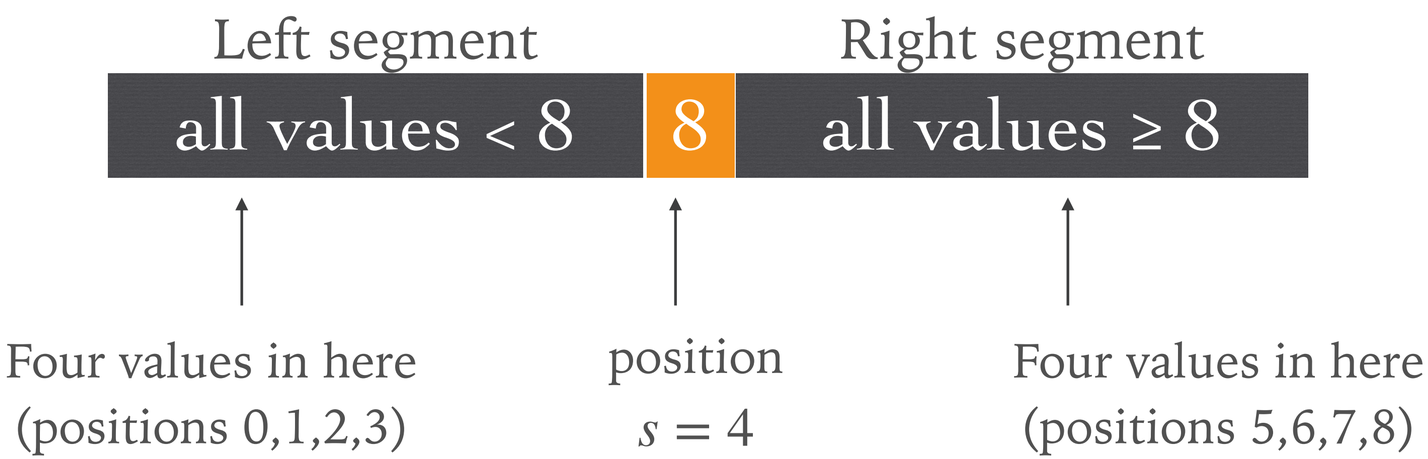
4 1 10 8 7 12 9 2 15



- The 5th-order statistic of the original list must be in the _____ segment

Choose "8" as the pivot element to partition the list into three segments:

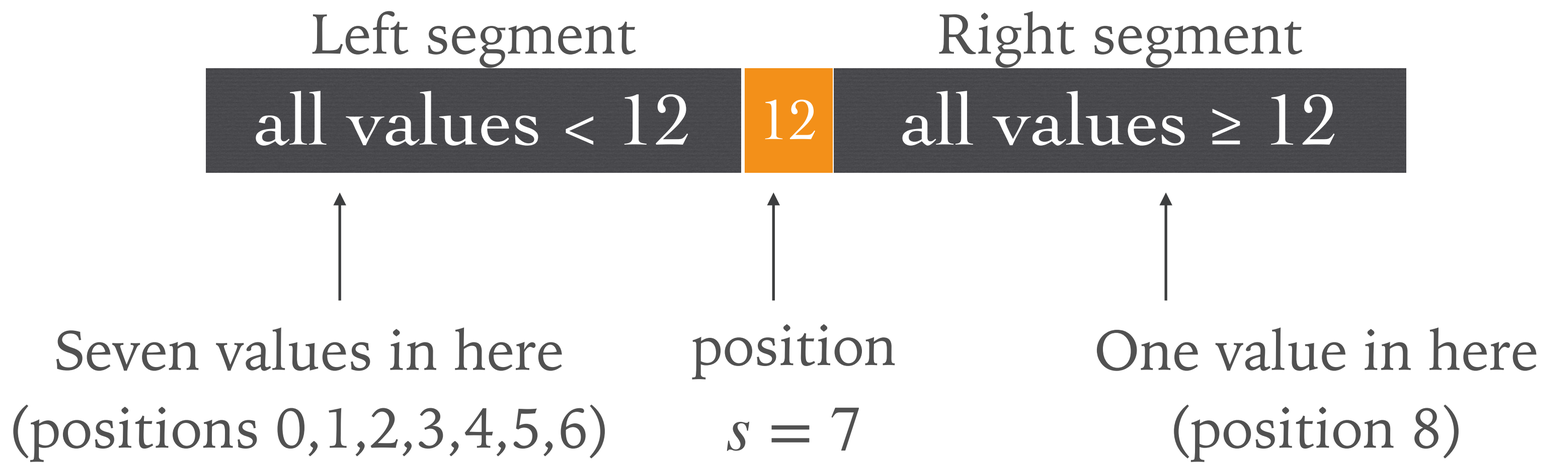




- The 5th-order statistic of the original list must be _____

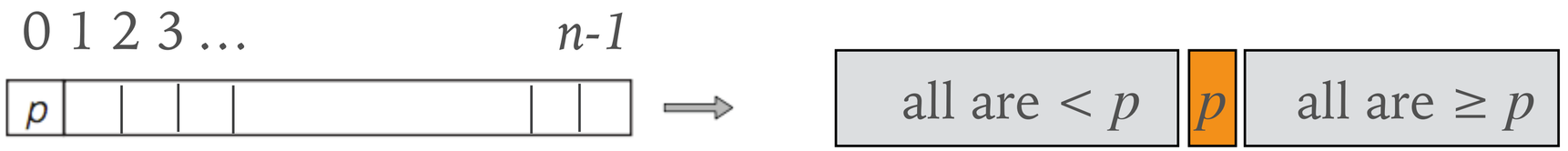
Choose "12" as the pivot element to partition the list into three segments:





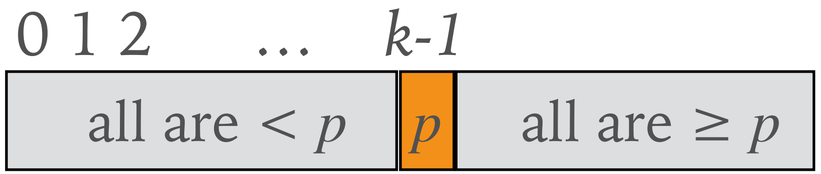
- The 5th-order statistic of the original list must be in the _____ segment.

Suppose we can partition an original list in three segments based on a pivot value p shown below:

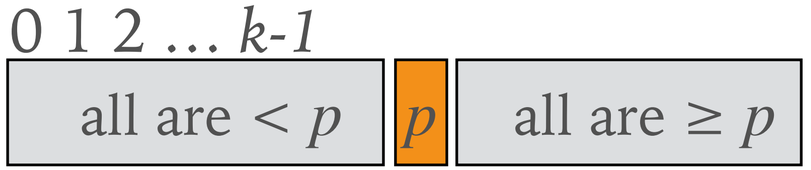


Three scenarios can happen:

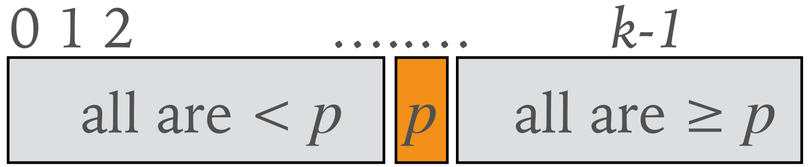
p is at position $s = k - 1$



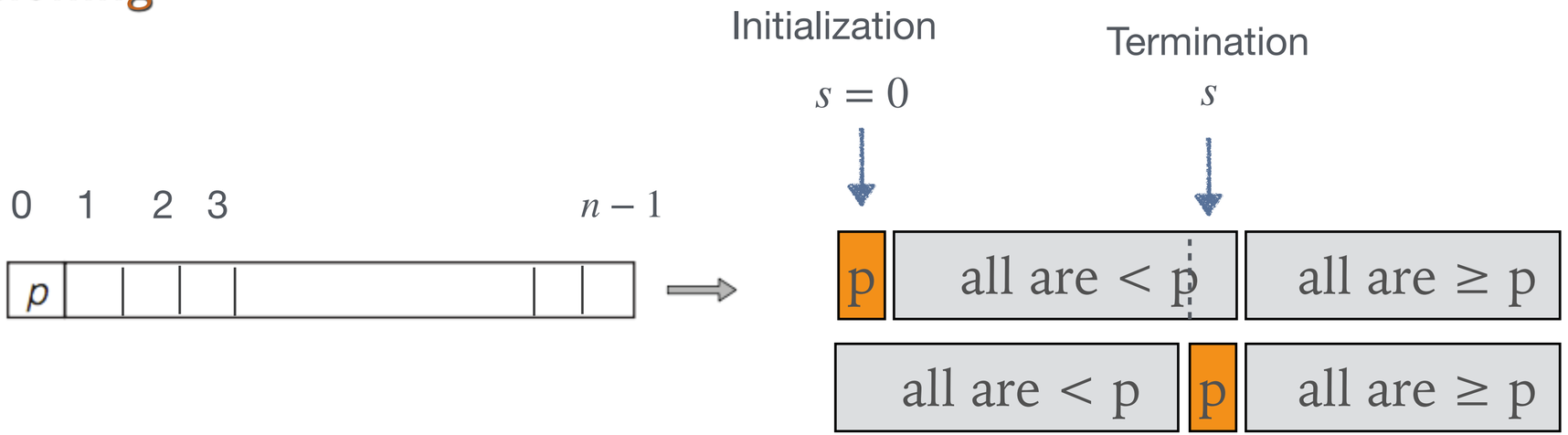
p is in position $s > k - 1$



p is in position $s < k - 1$



Lomuto Partitioning



Initialization

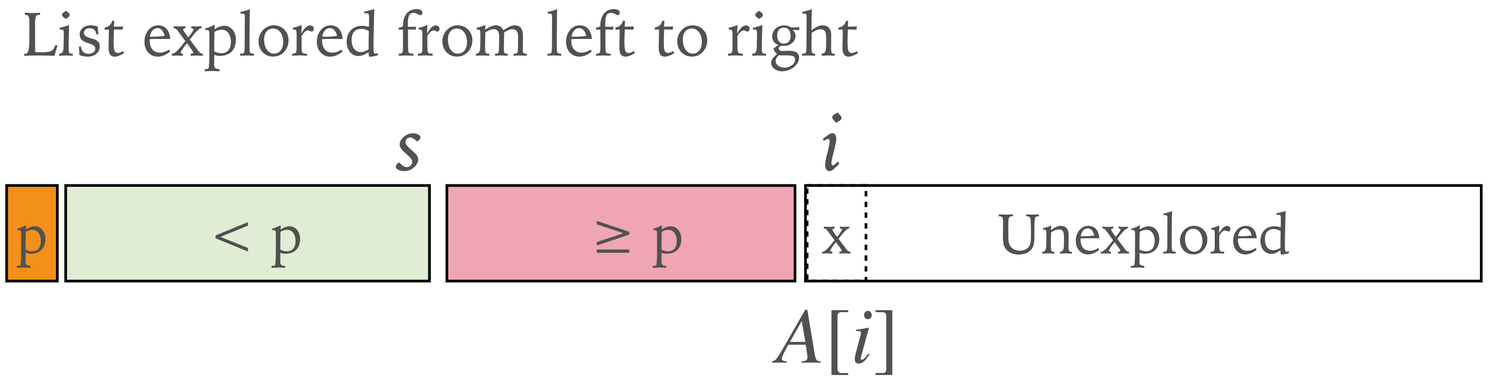
List index

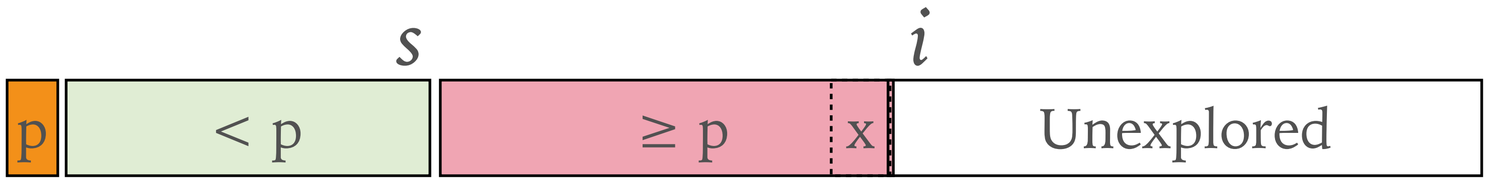
s i

$A[\cdot]$



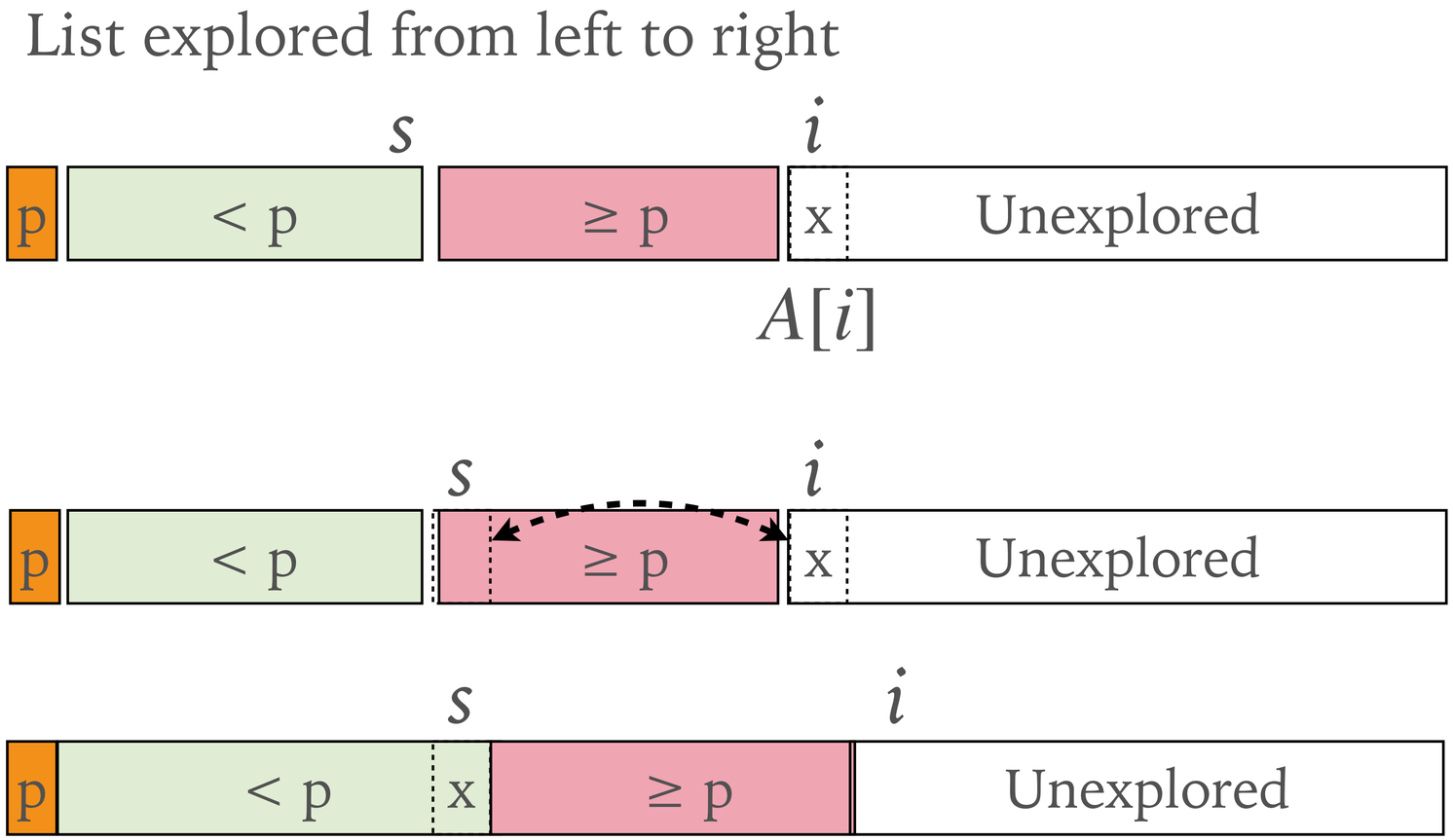
If $A[i] \geq p$, increment i, which will expand the $\geq p$ segment.



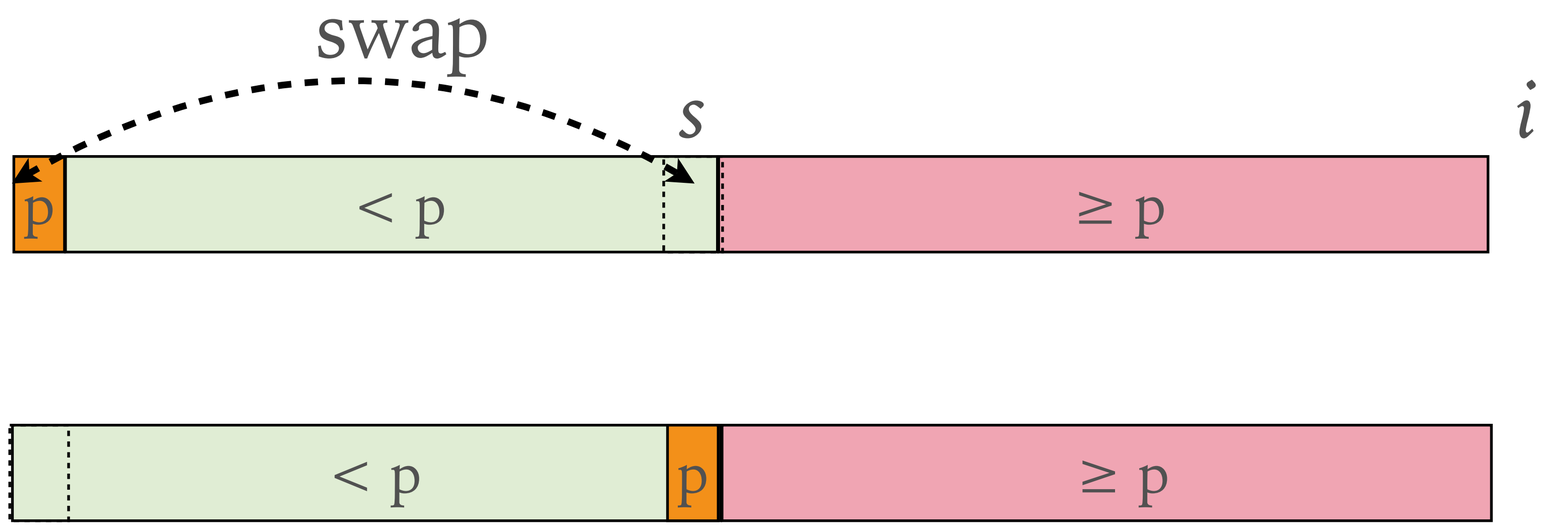


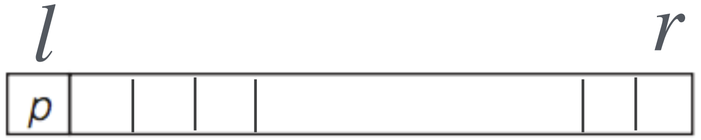
If $A[i] < p$,

- Increment s
- Swap $A[s]$ and $A[i]$, which will expand the $< p$ segment.
- Increment i



When all the elements have been explored,





Algorithm LomutoPartition($A[l..r]$)

Input: Subarray $A[l..r]$

Output: Partition of $A[l..r]$ and the new pivot position

$p \leftarrow A[l]$ Pivot element

$s \leftarrow l$ Initialize the index for left segment

for $i \leftarrow l + 1$ to r **do**

  if $A[i] < p$ **then** Basic operation

    $s \leftarrow s + 1$

    swap($A[s], A[i]$)

swap($A[l], A[s]$)

return s

Find the 5th order statistic

$l = 0,\ r = 8$

0	1	2	3	4	5	6	7	8
s	*i*							
4	1	10	8	7	12	9	2	15
4								
4								
4								
4								
4								
2								

0	1	2	3	4	5	6	7	8
			s	*i*				
			8	7	12	9	10	15

Assume the index starts at zero.

Algorithm QuickSelect($A[l..r]$,k)

Input: Subarray $A[l..r]$ and integer k $(1 \leq k \leq r - l + 1)$
Output: k^{th}-smallest element in $A[l..r]$

$s \leftarrow$ LomutoPartition($A[l..r]$)
if $s == k - 1$ **then**
    return $A[s]$
else if $s > k - 1$ **then**
    QuickSelect($A[l \,..\, s{-}1], k$)
else
    QuickSelect($A[s{+}1 \,..\, r], k$)

Efficiency of QuickSelect

Partition an n-element array requires $n - 1$ key comparisons.

- Best case if the split solves the problem or $C_{best}(n) = n - 1 \in \Theta(n)$
- Worst-case if $k = n$ and the array is strictly increasing
- Ex: Finding 9th-order statistic in {1,2,4,7,8,9,10,12,15} needs $n - 1$ partitions

$$C_{worst}(n) = (n-1) + (n-2) + \cdots + 1$$

$$= \frac{n(n-1)}{2} \in \Theta(n^2)$$

$$C_{avg}(n) \in \Theta(n)$$ (by using randomized pivot element)

For a list containing lots of duplicates, the three-way partitioning algorithm is recommended.

In-Class Assignment L3-1: Quick Select (Group of 2 to 3)

Summary

Quickselect algorithm selects top-k or bottom-k elements in a list.

Require no sorting.

Worst-case is $\Theta(n^2)$ but best-case and average-case are $\Theta(n)$.